



SRI VENKATESWARA COLLEGE
NEP: UGCF 2022



Dissertation Report
2025-26

Semester VIII

Global Readiness for Future Pandemics: A Comparative Analysis

Submitted by:

Lakshita Singh (0522008)

Under the guidance of

Dr. Sonakshi Jain
Assistant Professor
Department of Economics
Sri Venkateswara College

Sri Venkateswara College
University of Delhi
Benito Juarez Road, Dhaula Kuan
New Delhi -110021

Certificate of Originality

This is to certify that the Dissertation titled “Global Readiness for Future Pandemics: A Comparative Analysis” was conducted in the Department of Economics, Sri Venkateswara College, University of Delhi. The work embodied in this project is original and has not been submitted in part or in full, to this or any other University for any other degree or diploma. To the best of my knowledge and belief, it does not contain any published or written material from other sources, except cited work references.

No generative artificial intelligence technologies were used for the report writing/ only the authorized aids were used, which included suggestions from the supervisor regarding language and content. Generative artificial intelligence technologies if used were solely done in consultation with the supervisor. All sources have been properly acknowledged, including any ideas, extracts, or information taken from other sources. The work is free from plagiarism and no copyright laws have been violated to the best of my belief and knowledge.

Date: 8/05/2026

Signature of Student

Signature of Supervisor

Signature of Co-Supervisor

Acknowledgement

I would like to extend my deepest gratitude to Dr. Sonakshi Jain, Assistant Professor, Department of Economics, Sri Venkateswara College, University of Delhi, for her invaluable supervision, insightful feedback, and unwavering encouragement throughout this research journey. Her rigorous academic guidance and thoughtful suggestions have been instrumental in shaping both the direction and quality of this study.

I also express profound appreciation to Dr. Kanika Pathania, Department of Economics, Sri Venkateswara College, my ACR member, whose constructive criticism, methodological expertise, and generous support have significantly strengthened the analysis and clarity of the findings.

Finally, I thank the Department of Economics, Sri Venkateswara College, University of Delhi, for fostering an intellectually stimulating environment and providing the resources essential for completing this work.

TABLE OF CONTENTS

S.No	Topic	Page No.
1.	Introduction	5-11
2.	Research Objectives and Questions	11-12
3.	Literature Review	13-20
4.	Methodology	21-23
5.	Data Sources and Analysis	24-25
6.	Results	26-36
7.	Discussion	37-41
8.	Policy Recommendations	42-44
9.	Further Scope of Study	45-46
10.	Conclusion	46-48
11.	References	48-51
12.	Appendix	52-53

1. Introduction

1.1 Background

The COVID-19 pandemic, caused by the SARS-CoV-2 virus, created an unprecedented situation globally. It spread to several parts of Asia and then all over the world starting early 2020. The World Health Organization (WHO) declared the COVID-19 outbreak as a global health emergency on 30 January 2020 and as having turned into a global pandemic on 11 March. The WHO declared that the global health emergency brought about by COVID-19 was over in May 2023.

The coronavirus pandemic resulted in immense societal and economic chaos worldwide, resulting in the worst global recession since the Great Depression. Food and other supply shortages occurred because of disruptions in the supply chains and stockpiling. Human activity reduced so significantly that there was a remarkable decrease in pollution levels, which is unparalleled in recorded history.

Educational facilities and public spaces were either temporarily or permanently shut down in numerous places, and most planned events were either canceled or postponed in 2020 and 2021. Remote working gained popularity amongst office workers as the pandemic progressed. False information spread on social and sometimes mainstream media, and there was an escalation in political conflicts.

There was no specific drug treatment or cure to this viral disease in the first two years of the pandemic. It was only in 2021 that EMA CHMP endorsed the use of Paxlovid in treating infected adult patients in 2021, followed by FDA's EUA. Infections with COVID-19 were usually mild. In cases like this, supportive therapy involves using medication like paracetamol or nonsteroidal anti-inflammatory drugs (NSAIDs) for symptom management (body ache, cough), taking plenty of oral fluid intake, and having ample rest.

1.2 Worldwide Impact

The pandemic and responses to it damaged the global economy. On 27 February 2020, worries about the outbreak crushed US stock indexes, which posted their sharpest falls since 2008. Tourism collapsed due to travel restrictions, closing of public places including travel attractions, and advice of governments against travel. Job losses exceeded hundreds of millions of jobs, with over 40 million job losses in the US alone.

In China, multiple provincial-level officials of the Chinese Communist Party (CCP) were sacked over their response to quarantine regulations. Some people suggested that such moves were made in order to protect the position of CCP General Secretary Xi Jinping. The intelligence agencies of the US alleged that China deliberately suppressed statistics regarding cases of COVID-19. According to the Chinese authorities, China took swift and open actions.

The pandemic sparked panic, uncertainty, and pessimistic views in Brazil. The head of state showed utter disregard for WHO recommended mitigation methods, downplaying the risks of the coronavirus, increasing economic activity, misinformation on mask use, vaccines, and public health practices, and promotion of experimental treatment methods like hydroxychloroquine and ivermectin.

The outbreak of COVID-19 in the US sparked discussions and raised demands for implementing social policies characteristic of other developed nations, namely universal health care, universal child care, paid sick leave, and greater investment into public health. There were also protests against vaccine mandates in the United States.

The Iranian government was greatly impacted by the virus, as it had infected more than twenty-four parliamentarians and other influential individuals. The Iranian president, Hassan Rouhani, sent an open letter addressed to all world leaders on March 14, 2020, requesting assistance due to the absence of global market access.

Japan-South Korea diplomatic relations deteriorated when South Korea accused Japan of being "ambiguously and passively quarantining" its citizens following its announcement that citizens from South Korea must self-quarantine for two weeks.

Activists in countries such as Egypt, Turkey, and Thailand had been detained for disseminating false information about the virus.

1.3 Gaps in Pandemic Preparedness

The COVID-19 pandemic showed that even countries with relatively strong healthcare performed poorly when there was a sudden surge in cases, which is usually the case with pandemics. Hospitals reached their maximum capacity very quickly, and there were extreme shortages of hospital beds, ICUs, and other important medical facilities. Therefore, the surge capacity of healthcare infrastructure, defined as the ability of a health system to expand its operations quickly and efficiently in response to increased demand, failed miserably in most regions. For improvement in surge capacity it would require that existing facilities be used efficiently, professionals be trained to repurpose facilities during overwhelmingly high cases.

There were extreme inequalities among countries during the pandemic, the most apparent one being the unequal distribution of medical resources, including vaccines, testing kits, and treatment facilities. Some of the high income countries were easily able to acquire a large amount of vaccines in a short matter of time whereas others struggled to match up with them, experiencing delays in treatment , further increasing the speed of recovery.

Countries with prevalence of informal and hidden unemployment, were more vulnerable than others due to public measures like mandatory quarantining and forced lockdown. In developing countries mostly, a large segment of this vulnerable population relied on daily wages and they were robbed of that in an instance without any social protection, making their sustenance difficult.

Some of the major pandemic preparedness indices are namely, **International Health Regulations (IHR)** and **Global Health Security Index (GHSI)**. At the 64th World Health Assembly, a review of IHR (2005), specific to the H1N1 pandemic of 2009 was presented . The review found that -

“The world is ill prepared to respond to a severe influenza pandemic or to any similarly global, sustained and threatening public-health emergency”

While disease outbreaks and other acute public health risks and events are often unpredictable and require a range of responses, the International Health Regulations (2005) (IHR) provide an overarching legal framework that defines countries' rights and obligations in managing public health risks, events and emergencies that have the potential to cross borders. It includes 196 countries including WHO member states and follows a self-reporting mechanism to assess and detect public health risks. IHR does not rank countries or give out scores, rather it functions as a guiding body telling countries what to do and what not to do regarding strengthening preparedness, insisting compliance of their released guidelines.

Global Health Security Index (GHSI) benchmarks a total of 195 countries after assessing publicly available data regarding six categories, prevention, detection, reporting, and health system strength. The aim of this index is to strengthen preparedness across the globe and rank the countries accordingly. It enables cross-country comparison and is a trusted source of information by several researchers and policymakers.

Both of these indices, in their own way, provide useful data on pandemic preparedness, however they suffer from limitations as well. IHR mostly depends on self-reported data by the countries themselves without any external third party which means the data could easily suffer from bias. Also, GHSI does not show how effectively countries implement policies during an actual crisis.

Pandemics are inherently global in nature, requiring coordinated action across countries. However, during COVID-19, there were multiple instances of failure in coordination and cooperation among the countries. Even the implementation of the IHR framework and compliance was a challenge since individual countries did not show willingness to follow the recommended measures for containing the spread of virus.

Distribution of vaccines is another example of coordination failure, as mentioned previously as well. The supply chain was completely disrupted with many restrictions on import and exports creating shortages in not so developed regions. Even though pandemics are global in nature, there was a lack of coordinated effort against it by health systems around the world.

1.4 Significance of the Study

This pandemic uncovered vulnerabilities in the health system, governance structures, social and economic framework whilst also validating the robust nature of resilience systems across the world. The pandemic had varying impacts on countries, wherein, some were able to contain the virus and maintained low mortality rates, whereas others faced overwhelmingly high death rates due to failed healthcare systems, poor social structures, and less governance efficacy.

There were extreme disparities in terms of mortality rates, healthcare strain, economic disruption and societal resilience which highlights the differing levels of pandemic preparedness among countries. This paper aims at emphasising on the very aspect of how nations dealt with the pandemic earlier and whether they learnt their lesson and consequently took steps to strengthen their systems or not.

The differences in pandemic outcomes were not only due to varying healthcare infrastructure and medical facilities such as hospital bed availability, intensive care unit (ICU) capacity, and availability of medical personnel etc. but also a consequence of socioeconomic factors, governance quality, public trust in authorities, and capacity to mobilize resources rapidly. The complexity of pandemic preparedness can also be understood by looking at measures such as Epidemic Preparedness Index (EPI) and Global Health Security (GHS) Index which provide quantitative measures of countries' ability to respond to infectious threats, yet have shown varying predictive power during COVID-19.

As the risk of future pandemics is continuously increasing, it is a matter of global health concern to understand the factors that play an essential role in determining pandemic preparedness and response effectiveness. Thus, assessing which countries are best equipped to manage such health emergencies and why, can help inform better international health policies, resource allocation, and national preparedness strategies.

This study quantitatively analyzes the COVID-19 data, to find factors which best predict cross-country differences in excess all-cause mortality during that period (2020-21). It then uses the derived empirically validated relationships to construct a Pandemic Resilience Score. The traditional preparedness indices like EPI performed poorly during the crisis, however, this approach puts to use actual pandemic outcomes to identify the most influential predictors be it structural or social like health expenditure, digital connectivity, trust in governance etc. and applies them to current (2025) data of these countries. The study consequently ranks nations by their resilience score i.e. their predicted readiness for a future COVID-like pandemic.

The differences in what countries experienced globally during COVID-19 shows that preparedness is multidimensional, requiring comprehensive evaluation of all the structural and social aspects. Proper assessment of healthcare infrastructure, public trust and opinions, method of resource allocation, preparedness strategies could give way to effective policy implementation. This study aims to build on lessons learnt during the COVID-19 pandemic period by analysing the preparedness of total 136 countries and inferring how they would perform in the current time period if a similar magnitude of a pandemic were to happen again.

Most of the existing research on pandemic preparedness focuses either on aspects of healthcare capacity or governmental policies rather than incorporating multiple dimensions of factors into a single unified dataset. Thus, there remain critical research gaps which will be addressed in this study. Also, we know that many preparedness indices did not align well with actual COVID-19 outcomes, leaving unexplored links between perceived preparedness and pandemic performance. This distinction will be further looked at in the study.

The research question guiding this study is:

"If a COVID-19-like pandemic were to happen again, how would countries deal with it now(2025) in comparison to earlier(2020)?"

This study addresses gaps by placing an integrative focus on several key variables to comparatively analyze countries' readiness and response effectiveness, including both health infrastructure and socioeconomic factors.

Beyond identifying predictive factors, this research uniquely bridges the temporal gap between past performance and future readiness by constructing a Pandemic Resilience Score that applies empirically validated COVID-19 relationships to 2025 structural data. This forward-looking approach enables the formation of country rankings reflecting current capacity improvements (e.g., digital infrastructure gains, governance reforms) since 2020, offering a practical tool for real-time global health security monitoring that traditional static indices cannot provide.

The study's findings will contribute valuable insights to global health security discourse, help refine preparedness indices, and support policymaking aimed at future pandemic resilience.

2. Research Objectives and Questions

The primary purpose of this study is to analyse pandemic preparedness and readiness across countries and use a forward looking model to predict their performance in the current time period based on a multidimensional set of variables.

2.1 Research Objectives

This study aims to fulfill the following objectives by the end of the research process.

- To compare levels of pandemic preparedness and ability to respond effectively across all selected countries using a multidimensional dataset.
- To identify key predictors of COVID-19-like pandemic outcomes, with focus on both structural and social factors.

- To examine and understand the correlation between perceived preparedness and actual COVID-19 performance, based on country wise data.
- To provide policy-relevant recommendations for bolstering national readiness for future pandemics.

2.2 Research Questions

The research questions that will be answered during the process of this proposed research is as follows:

Q1) Which country-level structural and social factors are statistically significant predictors of excess all-cause mortality during the COVID-19 pandemic?

Q2) How does the interaction between trust in government and digital access influence public compliance with non-pharmaceutical interventions, and how is this reflected in excess mortality outcomes across countries?

Q3) Which countries emerge as most and least resilient to a future COVID-19-like pandemic based on the Pandemic Resilience Score?

Q4) What policy recommendations can be derived from the observed relationships between social determinants, digital access, health system financing, and excess mortality rates?

3. Literature Review

3.1 Rising risk of Pandemics

The COVID-19 pandemic was not a singular event but rather a result of a systematically increasing risk of global health crises. Marco et. al (2021) estimates the yearly probability of a COVID-scale pandemic is about 2% based on four centuries of outbreak data, implying an approximate 38% lifetime probability. He further mentions that this annual risk could increase up to threefold in coming decades due to consistently increasing emergence rates.

Pathogens can grow from a disease outbreak to pandemic proportions depending on the balance of a pathogen's characteristics, like transmissibility, case fatality rate, replication rate, and mutability, among other factors. Until the COVID-19 pandemic caused by the virus SARS-CoV-2, influenza was predicted to be the most likely pathogen to cause the next pandemic due to its high rate of mutation and transmission. However, that was not the case and we were faced with a new variant of a virus that researchers were not ready for.

While it is difficult to predict the type and magnitude of the next pandemic pathogen threat, making reasonable assumptions today may enable a more effective response in the future. Thus, in this study we assume that the next pandemic pathogen will have similar characteristics to the SARS-CoV2 virus.

3.2 Pandemic Preparedness and Pandemic Response

Rieger and Wang (2021) found that countries with high scores on pandemic preparedness metrics did not necessarily achieve the best outcomes which suggests that response systems, particularly the efficiency and appropriateness of government action and public compliance were equally or even more critical than pre-existing infrastructure. Countries like the U.S. spend a major chunk of their GDP on healthcare still they could not contain the virus in cognizance with their perceived preparedness. This literature distinguishes between pandemic preparedness—the extent to which systems have invested in advanced infrastructure, training, and coordination, and pandemic response—the actual policies and actions taken once a

pandemic emerges. This distinction is extremely important to understand that preparedness alone is insufficient. The COVID-19 pandemic demonstrated the same phenomenon.

This distinction has profound implications for this proposed study. While conventional pandemic preparedness indices emphasise more on infrastructure readiness, a predictive model of pandemic resilience must incorporate both the structural capacity to respond and the institutional and social conditions that enable rapid, effective implementation of that response.

3.3 Pandemic Preparedness Indices

Most research work done up until now has focused on preparedness indices like International Health Regulations (IHR) and the Global Health Security Index (GHSI) as measures of readiness against future pandemics, however there are studies that show that these instruments have a great mismatch between what is reported in these frameworks and how countries actually performed during the pandemic.

A study by Kachali et. al states that, “at country level, that the health-related outcomes from the first wave of the pandemic were primarily negatively correlated with the expected preparedness.” This study analysed the first 60 days after the date of a country's first COVID-19 related death in 2020 and compared it to the expected preparedness rank according to the existing global preparedness indices (IHR and GHSI) on a country level.

They found that for COVID-19 pandemic, national level health preparedness rankings were not an indicator of how well a country handled the pandemic. This is because pandemic outcomes were negatively associated with expected level of preparedness among countries which basically implies that countries who were perceived to be prepared better did not perform well during the first wave, measured by the death numbers.

The figure below from the study shows the scatter plot for GHSI national rank against mortality data. This scatter plot shows a slight indication that the better ranked countries seem to have higher mortality rates than the countries with lower ranks. However, it is important to note that the mortality data were extracted on June 1st and some of the regions had not yet peaked in terms of mortality at the time.

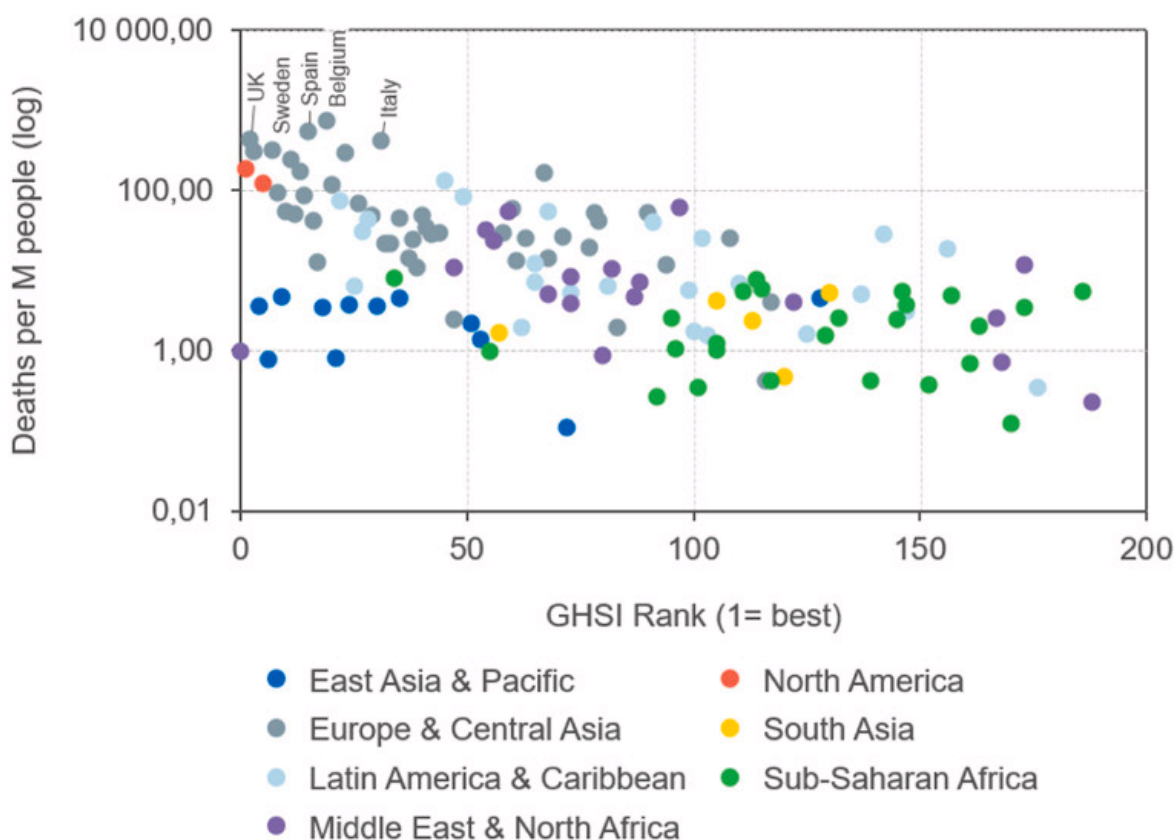


Figure 3.3 Note. Adapted from Kachali et al. (2020) . Figure plotted by the author.

The existing preparedness indices are constructed based on lessons learned from previous epidemics which had different characteristics i.e. some had lower transmission rates, lower mortality etc. These indices might not have been updated for today;s day and age especially taking into consideration air transmission of pathogens.

3.4 Determinants of Pandemic Outcomes: Beyond Health Expenditure

The OECD data presents a counterintuitive finding from pandemic outcome analyses that health expenditure as a percentage of GDP does not show a simple linear relationship with COVID-19 mortality. The United States spent 16.6% of GDP on healthcare, and still experienced among the highest per-capita mortality rates globally. China and Costa Rica, by

contrast, spent around 5.59% and 7.82% of GDP on healthcare and achieved significantly lower mortality(refer to figure-2 in appendix).

This suggests that what countries spend on and how they structure that spending matters more than just the volume of spending. Germany's superior performance was a result of many factors like them having a decentralized laboratory network that enabled rapid, widespread testing even in the early pandemic, a strong primary care infrastructure; and equitable access to healthcare regardless of income, which ensured that there was no delay in treatment due to financial barriers. Whereas, in the United States there was high spending on healthcare capacity but the testing infrastructure remained fragmented during the early days of the pandemic. They faced significant financial barriers to care for both uninsured and underinsured populations.

Western countries can possibly learn from practices from several East Asian countries regarding infrastructures, epidemiological surveillance and control strategies to mitigate the public health impact of the pandemic (Shokoohi et al., 2020). This is because there was contrast between the response system of nations like Vietnam and Taiwan and the the western nations who struggled to combat the overwhelmingly high mortality rates in their country. Western countries have invested in hospital-centered management of non-communicable diseases, however that clearly is not sufficient for pandemic preparedness.

"Financial and political investments in preparedness have been insufficient, and we are all paying the price." According to a report by GPMB, pre-COVID preparedness was stated as insufficient despite repeated warnings in 2019. Many governments view public health infrastructure as a cost center rather than an investment in national security and economic protection.

The pandemic exposed the miscalculation of preparedness by certain countries. The ones who were able to detect and contain early spread of virus due to consistent investment in rapid diagnostic capacity, information dissemination networks, and on training health professionals could prevent the exponential spread. Beyond just spending on health infrastructure, the proper development of quick response systems and implementation of non-pharmaceutical interventions was required to effectively mitigate the risks of the virus.

A study analyzed which policies, practices and responses amplified or undermined equity and social justice and identified opportunities for improved action. Ndumbe-Eyoh et. al(2021) states that, “current pandemic and emergency preparedness plans do not adequately account for the social and structural determinants of health and health equity.”

Rieger & Wang (2021), analysed responses from more than 100,000 individuals in 57 countries from March to April 2020. They argues that trust in government institutions depends on how citizens perceive the responsiveness of the policy. If the citizens believe that a measure is excessive or insufficient then they would show considerably less trust in government. Stronger stringency measures make people less likely to perceive government reactions as insufficient as shown in Figure-3.4(a), and more likely (not statistically significantly) to perceive the measures to be too extreme as shown in Figure-3.4(b).

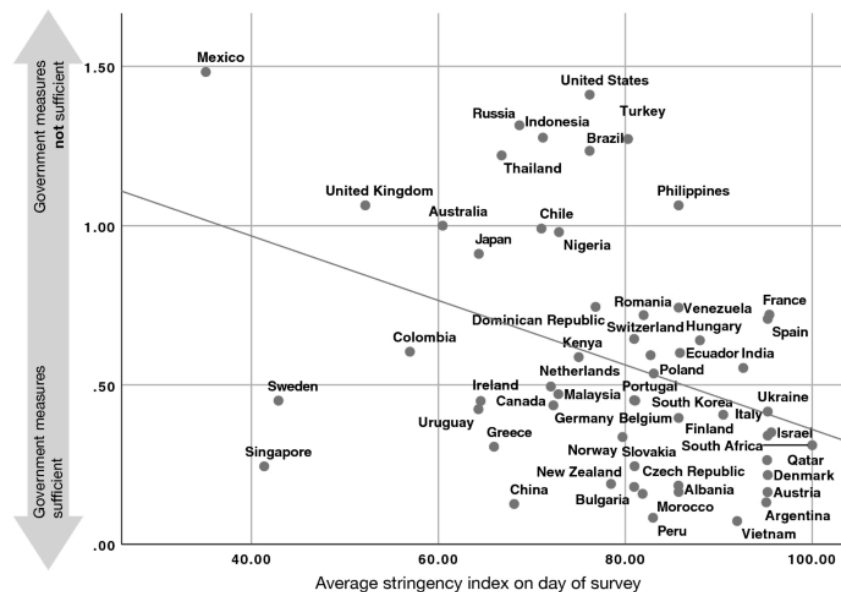


Figure 3.4(a) Average stringency on day of survey (x-axis) versus opinion of people in a country that the government measures are not sufficient (y-axis)

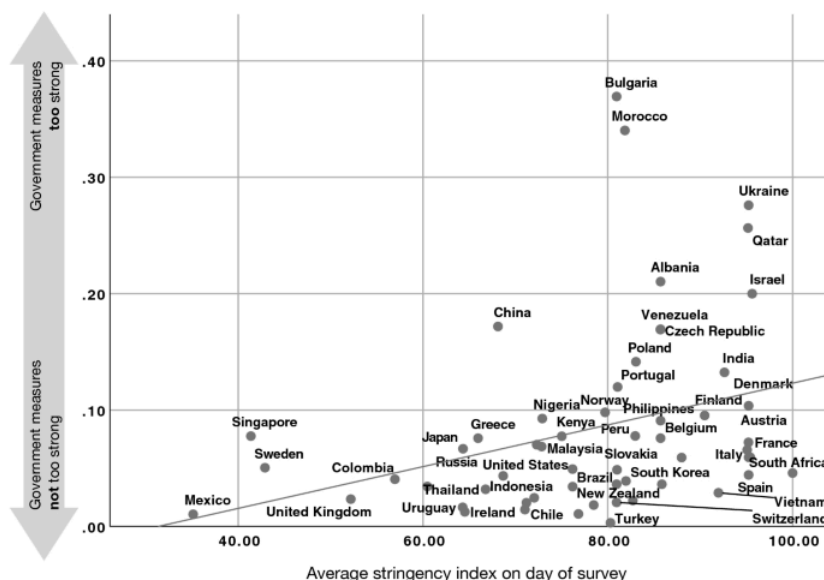


Figure 3.4(b) Average stringency on day of survey (x-axis) versus opinion of people in a country that the government measures are too extreme (y-axis)

The implications are profound. Governments which were perceived as trustworthy, responsive, and transparent, their citizens voluntarily complied with non-pharmaceutical interventions like wearing a mask, social distancing and following quarantine rule, resulting in reduced transmission rates. However, in countries where the governing body's image was ridiculed with considerably low trust, it required coercive actions to make citizens follow the laid down rules and regulations. (refer to figure-1 in Appendix).

They also found that conspiracy theory beliefs influenced citizens' perception greatly. Even when mortality rates were incredibly high, the government's response to it was labelled as too extreme without any objective backing for such claims. In structured surveys in Germany, conspiracy beliefs were significantly correlated with perceiving lockdowns as excessive but were not correlated with perceptions of insufficient response.

3.5 Comparative International Responses: East vs. West

The COVID-19 pandemic revealed striking differences in response effectiveness between East Asian and Western nations. Shokoohi et al. (2020) highlights that East Asian countries—South Korea, Vietnam, Taiwan, and Hong Kong—achieved substantially lower mortality despite comparable or earlier case exposure, through rapid implementation of community-centered strategies.

A significant commonality among these successes was the implementation of these strategies from the early days of spread, before cases had spread widely and overwhelmed contact tracing and isolation capacity. This window before exponential growth becomes overwhelming, was used strategically to prevent subsequent waves in those countries.

By contrast, Western nations largely missed this critical early window. Shokoohi et al. (2020) documented that the US and UK delayed strong response measures due to underestimation of the threat and reluctance to disrupt economies by political leaders, resulting in exponential community transmission before lockdowns and systematic testing were implemented. Italy, France, and Spain implemented lockdowns only weeks after their first cases. Sweden pursued a relaxed mitigation strategy with limited testing, resulting in high mortality.

The contrast between Germany and most other Western nations is instructive. Germany, despite early initial uncertainty, rapidly moved towards evidence-based response i.e. early expansion of testing capacity, clear communication with the public, and decision-making informed by scientific expertise.

3.6 Research Gaps

Despite extensive research on pandemic preparedness and COVID-19 outcomes, several significant gaps remain, which this dissertation addresses.

First, most existing literature conducted retrospective analyses of the 2019-2021 pandemic period, examining what factors explained outcomes during COVID-19. However, there is limited research constructing forward-looking predictive models that assess which countries are best positioned for future pandemics based on their current state (2024-2025). This study, on the other hand, constructs a Pandemic Resilience score to get an idea of a country's

performance in a future pandemic-like situation, since the risk of it seems to be consistently increasing.

Secondly, the dependent variable in this study is Excess Mortality Rate which will act as a much more reliable measure of pandemic outcomes. Since we conducted a cross country analysis, considering COVID-19 death rates as a measure could weaken the model as there is great variation in data collection and reporting standards as well as populations of these countries. Literature which used death numbers as a measure faced the same problem.

The last gap addressed in this study is that most existing research often examined either healthcare infrastructure (beds, expenditure, workforce) or social determinants (trust, inequality, education) separately. This study brings both aspects within a single unified framework, enabling evaluation of social predictors alongside healthcare indicators.

According to the literature findings, effective pandemic management needs much more than just medical interventions. Hospital beds and ventilators being not as important as social variables like trust, good governance, and prompt community-centered solutions. There is evidence that argues that social determinants might be equal or even more important than the structural ones, thus, eliminating them from this study would not be a smart choice.

We aim at creating a thorough predictive model that rates nations' preparedness for potential COVID-19-like pandemics, to fill in the above mentioned research gaps and provide answers to problems faced in the process of policy formulation and implementation.

4. Methodology

The study adopts a quantitative, cross-country research design built around excess all-cause mortality during the COVID-19 pandemic and country-level structural determinants, using only secondary data sources. A total of **136** countries have been included based on the availability of comparable data for excess mortality and the selected independent variables for the years 2020–2021.

The empirical analysis will focus on a historical outcome period covering the main COVID-19 shock, e.g. January 2020–December 2021, to define excess mortality and estimate the regression model. A “current” predictor period around 2023–2025 for covariates like trust in government, internet penetration, health spending and elderly population, which will be plugged into the model coefficients to generate a 2025 pandemic resilience score.

4.1 Dependent Variable

The study will consider the following variables:

- **Excess Mortality Rate-** It is the percentage difference between observed all-cause deaths during the COVID-19 pandemic (2020–2021) and expected deaths based on pre-pandemic (2015–2019) mortality trends, adjusted for seasonal patterns and long-term demographic changes. The rationale behind using this measure involves the fact that it captures the no. of deaths while also adjusting for data variations across countries.

4.2 Independent Variables

The five country-level indicators have been chosen keeping in mind, research objectives, data availability constraints and literature findings:

- **Health Expenditure (per capita terms)**- This variable measures the combined public and private health expenditure on all health goods and services, divided by the country's population. In comparison to other similar measures, this one accounts for variation in population of countries and the resource available per person.
- **Trust in Governance**- The proxy variable chosen for this measure is the Voice and Accountability Indicator which aggregates survey questions from the Gallup World Poll on citizens' political voice, confidence in institutions, and perceived accountability mechanisms. This particular proxy had data available for most number of countries and it encapsulates the effect well.
- **Internet Penetration Rate**- This refers to the percentage of population (15+) using the internet from any device or location, in the past 3 months. It reflects digital connectivity essential for pandemic communication and remote service.
- **Population Density**- This measure is simply the total population of the country divided by area of land i.e. people per square kilometer. Since the population variation had to be accounted for, logarithmic transformation is applied on this measure in this study.
- **Share of Elderly Population** - This variable states the percentage of population that is aged 65 years and above. It is an important demography indicator which measures population ageing.

4.3 Empirical Model Specification

The first step in the methodology applied in this study, involves estimating the relationship between the above mentioned determinants and pandemic outcomes, using a multivariable linear regression model with excess mortality rate as the outcome and the five country level

indicators as covariates. The model is based on 2020–2021 data to identify how these variables impacted pandemic performance during the COVID-19 period.

$$\text{Excess Mortality Rate}_i = \beta_0 + \beta_1 \text{Health Spending (\% GDP)}_i + \beta_2 \text{Trust in Governance}_i + \beta_3 \text{Internet Penetration Rate}_i + \beta_4 \text{Population Density}_i + \beta_5 \text{GDP per Capita}_i + \varepsilon_i$$

where i indexes countries, β_0 is the intercept, β_1 to β_6 are the coefficients to be estimated, and ε_i is the error term.

4.4 Construction of the Pandemic Resilience Score

The estimated regression coefficients (beta values) from this model are then used as weights to construct a composite Pandemic Resilience Score for each country. For this step, the values of the independent variables (based on latest available data i.e. 2024-25) are plugged into the weighted linear equation, so that countries with more favourable profiles on the significant predictors receive higher resilience scores, representing stronger predicted capacity to handle a future COVID-19-like pandemic.

In this study, Pandemic Resilience Score can be defined as a quantitative measure that indicates better adaptation, coping and resilience against pandemic shocks.

To improve comparability across countries, the raw scores obtained from the weighted equation are further **normalized using a min–max scaling technique**, transforming them into a standardized range from 0 to 100

$$X' = \frac{X - X_{min}}{X_{max} - X_{min}}$$

Under this transformation, the country with the lowest raw score is assigned a value of 0, representing the least resilience, while the country with the highest raw score is assigned a value of 100, representing the highest resilience. All other countries are scaled proportionately within this range.

5. Data Sources and Analysis

5.1 Data Sources

Data has been sourced from globally recognized repositories and surveys, including

- World Health Organization
- World Bank
- World Population Review
- Our World in Data
- Country-specific published statistics

5.2 Data Preparation

The process of forming a multidimensional dataset involved collecting data of all dependent and independent variables for the year 2020 and the latest available data year (2024/2025) for all 136 countries. It involved the use of various global data repositories for accessing the required secondary data.

Data was extracted from these sources and consolidated as a unified dataset in Microsoft Excel. If the data was mentioned in different formats or units, then appropriate conversions were applied to ensure consistency across variables. After extraction, the next step involved cleaning of the data. This included checking for missing values, inconsistencies, and duplicate entries. The aim here was to include as many countries in the dataset. The main constraint with data availability was with the social determinants like trust in governance. Since data for the same set of countries was required, the final number of countries got reduced to 136, as certain determinants had different countries missing.

Countries with significant data gaps across key variables were excluded from the final sample to maintain the robustness of the analysis. Outliers were identified through preliminary statistical checks and visual inspection in R , and their potential impact on the analysis was assessed before deciding whether to retain or exclude them.

To prepare the data for the first step which was the Multivariable regression analysis, logarithmic transformations were applied to variables like health expenditure per capita and population density. This ensures interpretation in percentage terms which is sometimes essential in cross country analysis as there is extremely varied data. The data for 2020 and 2025 were also separated into two excel files for regression purposes.

The rationale behind logarithmic transformation being applied to variables like, health expenditure per capita and population density, is to improve the statistical properties of the data and enhance the robustness of the regression analysis. Since we have a dataset with multiple countries there is bound to be variation with certain countries having extremely large values as compared to others. To reduce the said skewness of the distribution, log values of these variables were used, which compressed the scale of large values and expanded smaller values for an equitable analysis.

One of the key assumptions of ordinary least squares (OLS) regression is heteroscedasticity which also gets solved by log transformation. Variance of the error terms increases with the magnitude of the independent variables so in their raw form, variables such as population density and health expenditure exhibit non-constant variance, which violates that assumption.

After cleaning and transforming, the final Excel file was uploaded in R (statistical software) for further analysis. Final checks were made to ensure the file could be used appropriately in the software.

6. Results

The results from the multivariable regression model reveal the significant determinants of excess mortality during the pandemic period. The model incorporates five key factors-log of health expenditure per capita, trust in governance, internet penetration, proportion of elderly population (aged 65 and above), and log of population density, and captures their relative influence on the outcome variable. This first step of the methodology provides an understanding of what variables are important and whether they should be included as determinants in the construction of Pandemic Resilience Score.

6.1 Summary of Regression Results

Variables	Coefficient	t-value
Constant	0.005419***	5.484
log(HExp)	-0.000645*	-2.242
TIG	-0.003903*	-2.162
IP	0.0000318*	2.482
EldP	0.0002168***	4.995
log(PD)	-0.0003915**	-3.148

*p<0.05; **p<0.01; ***p<0.001

Figure 6.1 OLS Regression Results(2020)

The estimated coefficients derived from regression analysis provide a mix of expected and nuanced relationships between dependent and independent variables.

The value of log of health expenditure per capita coefficient is -0.000645 which implies a negative association with excess mortality rate. A 1% increase in health expenditure per capita would lead to a reduction in excess mortality of approximately 0.00000645 units. This particular result aligns with the general expectation that higher spending on healthcare infrastructure improves system capacity, provides quicker access to medical services, and enables more effective responses to health emergencies, consequently reducing mortality rates.

This means countries with higher levels of health expenditure should be able to combat pandemic-like situations better, however, the relatively low magnitude of the estimated coefficient proves that increased spending alone is not sufficient for substantially improving pandemic outcomes, especially in the short run. Health expenditure needs to be accompanied by effective resource allocation and proper usage of funds to provide better-equipped hospitals, more healthcare personnel, and stronger public health infrastructure for the maximum number of the population.

The next variable, trust in governance uses a proxy called the voice and accountability indicator. and sheds light on the importance of institutional quality in contributing to better outcomes. The estimated coefficient -0.003903 , signifies it as an important determinant of excess mortality. The coefficient for trust in governance is negative which implies that higher levels of trust in governance are associated with lower mortality rates in the country. A one-unit increase in this variable corresponds to a reduction in excess mortality of approximately 0.003903 units.

In countries, where citizens trust the governing body, there is likely to be greater compliance with non-pharmaceutical interventions such as social distancing, mask-wearing, and lockdown rules and regulations. This quality ensures that measures are taken seriously by the citizens, facilitating effective communication and reducing chances of misinformation being spread by anti-social elements. Since the magnitude of this variable is comparatively larger it indicates that government capacity is essential not only in policy formulation but also adherence and implementation of strategies.

In contrast, internet penetration rate shows a small but positive association with excess mortality, with a coefficient of $+0.0000318$ i.e. A one percentage point increase in internet penetration is associated with an increase in excess mortality of approximately 0.0000318 units. This result can be explained by considering that high internet penetration rates improve the process of data collection and consequent reporting of mortality rates. This leads to accurate measurement of deaths and the small magnitude of the positive coefficient reflects better reporting rather than an actual increase in mortality.

Also, countries with higher internet penetration are more urban and globally integrated, which may facilitate faster transmission of infectious diseases due to higher mobility, population density, and international connectivity. Therefore, the observed relationship may be capturing underlying structural characteristics.

Another variable with a positive estimated coefficient is the percent of elderly population(65+) in the country. The results reveal that this determinant is the strongest predictor out of all five of the variables, with a high magnitude of $+0.0002168$. This implies that a one percentage point increase in the share of individuals aged 65 and above is associated with an increase in excess mortality of approximately 0.0002168 units.

Older populations are clearly more vulnerable to severe outcomes from infectious diseases and this is due weaker immune systems, higher prevalence of chronic conditions, and increased susceptibility to complications among older individuals. Thus, demographic composition is an important determinant in understanding cross-country differences in pandemic outcomes.

The final variable i.e. log of population density is found to have a negative coefficient (-0.0003915) which indicates that higher population density is associated with lower excess mortality. The general expectation would be of a positive association between the two variables however this counterintuitive result, could just be a reflection of the fact that densely populated areas often have better healthcare infrastructure, greater availability of medical services, and more efficient dissemination of information which leads to reduction in mortality rates.

The Multiple R-squared value derived from the analysis is 0.2618 which means 26.18% of the variation in excess mortality across countries can be explained by the variables included in the model. The Adjusted R-squared, is slightly less i.e. 23.34% since it takes into consideration the number of variables included in the model. This suggests that after adjusting for model complexity, approximately 23.34% of the variation in excess mortality is explained by the regressors.

Here, overfitting is not a problem since the gap between the Multiple R-squared value and Adjusted R-squared value is very less.

The F-statistic of this model is 9.222 with 5 and 130 degrees of freedom, which is highly significant ($p = 1.56e-07$). This confirms that the set of independent variables, taken together, has a statistically significant relationship with excess mortality and that the model provides a substantially better fit than one with no explanatory variables.

Diagnostic checks further support the validity of the model. The residuals appear to be approximately normally distributed, with a median close to zero and symmetric first and third quartiles. This indicates that the normality assumption is reasonably satisfied. Additionally, the absence of major outliers suggests that the estimates are not unduly influenced by extreme observations, thereby enhancing the reliability of the results.

6.2 Pandemic Resilience Scores

The beta values obtained for each variable after performing regression acted as weights in the calculation of Pandemic Resilience Score for all countries included in the study.

The Pandemic Resilience Score was calculated using a weighted linear combination of the selected variables, where each country's observed value for a given variable was multiplied by its corresponding beta coefficient.

The final score equation is given as follows:

$$\text{PRS} = 0.005419 - 0.0006449(\log\text{HExp}) - 0.003903(\text{TIG}) + 0.00003179(\text{IPR}) - 0.0003915(\log\text{Pop. Dens}) + 0.0002168(\text{EldP})$$

where i indexes countries.

This is a forward looking model as it uses the current 2025 data of health expenditure per capita, trust in governance, internet penetration rate, share of elderly population and population density based on 2021 weights of these variables, thus providing proportionate significance to all the predictors. In doing so, the model assumes that the relationships identified during the pandemic period continue to hold in the present context.

For every country, data was entered into the equation to derive the score. However, since these raw scores are not directly comparable across countries due to differences in scale and units, a normalization procedure was applied to get the final pandemic resilience scores ranging from 0 to 100, with the least resilient country with score 0 and the most resilient country with the top score 100.

6.3 Ranking of Countries

The ranking of countries has been done based on the computed Pandemic Resilience Scores. These scores indicate the country's relative capacity to withstand pandemic shocks and face a potential global health crisis like Covid-19.

Post normalizing the scores, countries were ranked in descending order of their scores. The top ranking country with PRS of 100 represents the highest level of resilience based on the key variables taken into consideration in the study. Similarly, the country with the lowest score is placed at the bottom of the ranking. The rankings provide a clear metric of differences in pandemic resilience and preparedness across the selected set of countries

The following table shows the ranking of all 136 countries:

S.no.	Country	Pandemic Resilience Score
1	Denmark	100
2	Finland	96.90873356
3	Nicaragua	91.70055051

S.no.	Country	Pandemic Resilience Score
4	United States	91.17580099
5	Tanzania	91.05983648
6	Italy	90.95371947
7	New Zealand	90.45267814
8	Estonia	89.04195816
9	Australia	88.40058869
10	Canada	86.92060046
11	Luxembourg	86.71206317
12	Belgium	85.68309085
13	Ghana	84.93312118
14	Madagascar	84.75903628
15	France	81.07541655
16	Uruguay	80.05932789
17	Pakistan	78.38244635
18	Lebanon	78.20463952
19	Uzbekistan	77.91633372
20	Czechia	77.23824501
21	Kosovo	76.71197375
22	Slovenia	75.60587592
23	Jordan	75.05476409
24	Chile	72.87030668
25	Costa Rica	72.49023924
26	Austria	72.45860017
27	Hungary	71.2218621
28	Sri Lanka	70.40918864
29	South Africa	67.01615071
30	Portugal	65.89883116
31	Jamaica	65.23686939
32	Romania	61.55945557
33	Colombia	60.57587502

S.no.	Country	Pandemic Resilience Score
34	Cyprus	60.29288968
35	Spain	60.27876139
36	Germany	59.05721265
37	Grenada	58.85375524
38	Argentina	58.71676393
39	Russian Federation	57.91757422
40	Mauritania	57.46056594
41	Croatia	56.9465913
42	Bulgaria	56.65530184
43	United Arab Emirates	56.18700512
44	Brazil	55.8212669
45	Indonesia	55.00495046
46	Dominican Republic	54.44347627
47	Nepal	53.32286481
48	Japan	52.76642157
49	Barbados	52.62067261
50	Moldova	51.82183016
51	Paraguay	50.57520208
52	Kyrgyz Republic	50.13493831
53	Senegal	49.71101542
54	Philippines	49.31308237
55	Iran, Islamic Rep.	48.89473388
56	Peru	48.76102507
57	Netherlands	48.31577269
58	Mongolia	47.77730002
59	St. Kitts and Nevis	47.73585627
60	Antigua and Barbuda	47.06451608
61	Norway	46.6665954
62	Tunisia	46.33819057
63	Ecuador	45.71830445

S.no.	Country	Pandemic Resilience Score
64	Korea, Rep.	45.68976429
65	Mexico	44.78348103
66	Guinea	44.58159629
67	Greece	43.69924625
68	St. Vincent and the Grenadines	43.23980876
69	Bahamas, The	40.78401085
70	Dominica	40.4314466
71	Morocco	39.9189778
72	Latvia	39.66293462
73	Bosnia and Herzegovina	39.12004486
74	Poland	38.7589504
75	St. Lucia	38.28701163
76	Togo	38.0879708
77	Malta	37.91969265
78	Guatemala	37.85178543
79	Albania	37.11782623
80	Türkiye	36.75303146
81	Kenya	36.64368263
82	Lithuania	36.39899908
83	Botswana	35.37615339
84	Malaysia	35.24478797
85	Rwanda	34.44391319
86	India	34.36838599
87	Sudan	34.25528928
88	El Salvador	33.63612975
89	Malawi	33.41643079
90	Burkina Faso	33.37674543
91	Belize	33.37022496
92	Haiti	32.50212447
93	Serbia	32.09580574

S.no.	Country	Pandemic Resilience Score
94	Liberia	31.92745454
95	Bolivia	30.87258647
96	Zambia	30.8621899
97	Sierra Leone	30.82551685
98	Panama	30.60876461
99	Gambia, The	29.91869129
100	China	29.2136746
101	North Macedonia	29.09280327
102	Mozambique	27.52824842
103	Bangladesh	27.20821721
104	Uganda	26.21835273
105	Sweden	25.28201512
106	Thailand	24.54356384
107	Zimbabwe	24.4888845
108	Mali	24.29187082
109	Benin	23.71673573
110	Ukraine	23.36170064
111	Kazakhstan	23.14588595
112	Georgia	23.13942449
113	United Kingdom	21.79101154
114	Venezuela, RB	21.75606673
115	Côte d'Ivoire	20.056464
116	Angola	19.96690725
117	Algeria	18.99466781
118	Nigeria	18.96880747
119	Myanmar	18.10678722
120	Guyana	18.03039544
121	Honduras	16.8103212
122	Cameroon	16.28097745
123	Trinidad and Tobago	15.97708645

S.no.	Country	Pandemic Resilience Score
124	Afghanistan	14.56710833
125	Suriname	13.9064938
126	Congo, Dem. Rep.	13.84249734
127	Congo, Rep.	13.36319026
128	Belarus	11.6777541
129	Mauritius	10.90965755
130	Ireland	10.86025602
131	Ethiopia	9.498119584
132	Viet Nam	7.525246264
133	Niger	7.220632962
134	Namibia	5.99624379
135	Egypt, Arab Rep.	0.623003722
136	Cambodia	0

Table 6.3 Ranking of countries based on Pandemic Resilience Scores

At the top , we have **Denmark** as the most resilient country with a score of 100, followed closely by **Finland (96.91)**, **Nicaragua (91.70)**, and the **United States (91.18)**. The top quartile of countries has both developed and developing nations included. This means pandemic resilience is not singularly dependent on GDP or economic development but by an amalgamation of various different social and structural factors.

Other countries which are also ranked highly include **Italy**, **New Zealand**, **Australia**, and **Canada**. This is because these countries have well-built healthcare systems and proper governance frameworks. On the other hand, countries like Ghana and Madagascar are also relatively high in rankings showcasing that less elderly population and other social conditions can improve pandemic outcomes greatly.

Countries such as **Spain**, **Germany**, **Brazil**, and **Indonesia**, display moderate resilience and are ranked in the middle segment. In contrast, the lower end of the ranking is dominated by

countries like **Afghanistan (14.57)**, **Suriname (13.91)**, **Democratic Republic of Congo (13.84)**, and **Cambodia (0)** with weaker health systems, conflict ridden environment and less resources to allocate for combating a pandemic. These countries exhibit the lowest resilience scores. These rankings reflect a need for proper mitigation strategy and allocation of resources towards these states so that they are at least able to improve pandemic outcomes to some extent.

India ranks 86 out of 136 countries mainly due to less health expenditure per capita and a very high population density that cannot even be offset by other favourable conditions. Countries like the **United Kingdom** and **Sweden**, appear relatively lower in the ranking compared to perceived pandemic preparedness. This result solidifies the idea that high expenditure is not enough, it should also be accompanied by other social and structural factors.

The rankings take into consideration multiple important variables and the countries which showed balanced performance in all of them scored high resilience scores. Whereas, being extremely weak in any of the factors led to decrease in score dramatically and consequently lower rankings as well. It suggests that overall preparedness is a team effort by all the relevant variables that were and were not included in the study.

7. Discussion

7.1 Key Findings

It has been strongly established through existing literature, results of the regression analysis and the final ranking of countries that pandemic outcomes are never and will not be determined by any single component but through an interaction of various structural and social characteristics. While higher health expenditure and greater trust in governance lead to lower mortality, the demographic structure i.e. an ageing population and spread of misinformation become liability during a crisis. The interpretation of predictors like population density and internet penetration in a country remain complicated as it presents results showing varied effects.

The analysis of the regression model shows the R-squared value as 0.2618 which is relatively low predictive power for a model. However, in cross-country empirical analyses, particularly those involving complex phenomena such as pandemics, such levels of explanatory power are not uncommon due to the presence of unobserved heterogeneity, measurement differences, and institutional variations across countries.

A key conclusion from the analysis is the role of healthcare expenditure, which shows a negative association with excess mortality. This supports the widely accepted view in the literature that stronger health systems are fundamental to crisis response. The rationale behind taking health expenditure per capita instead of health expenditure as percentage of GDP was to derive a negative association between the two variables instead of a counterintuitive finding which would make things complicated.

The coefficient of trust in governance being significantly high provides a particularly important contribution to the discussion. It is the non-pharmaceutical interventions that require voluntary compliance and public measures rely heavily on the same. Trust becomes very crucial as it acts like social capital to help governing bodies implement their strategies effectively resulting in rapid positive impacts in pandemic outcomes i.e. reduced mortality rates. The overall pandemic preparedness of a country gets drastically affected by this predictor as it can amplify or weaken the effectiveness of interventions immensely.

The strongest predictor out of all turned out to be the share of elderly population in a country which reinforces the argument that an aging population is a critical risk factor during a pandemic. Its impact could amplify if the population density is extremely high as well. Demographic structure of a country, thus, has important implications for policy formulations. Steps like targeted interventions should involve prioritizing taking care of the elderly first and then moving in phases towards other demographics.

Certain countries with high penetration rates tend to have better data reporting techniques which results in capturing a higher number of deaths in comparison to countries which have flawed or rather broken reporting systems. This is reflected in the positive association established in regression analysis.

Interpretation of population density's role in pandemic preparedness, although seems straightforward, is a little more nuanced. This is because high population density areas are generally urban neighbourhoods that have much better healthcare available along with public measures being applied here with consistent monitoring. Thus, whether in a high population density area, mortality reduces or increases depends on other conditions being in favour or against the betterment of citizens. Thus, the interaction effect of variables might have to be studied in further research.

The rankings as a whole give a good enough picture of what strength of a country makes them go up the rankings and vice-versa however that does not mean that we take the ranking of each country as absolute. For example, Pakistan ranks higher than countries like South Korea and Sweden, which in reality is not at all true with regards to pandemic preparedness. This ranking is a result of misreporting and underreporting mortality rates along with population density working here in Pakistan's favour implying better pandemic outcomes even though that is not the case since their dense areas are not properly managed.

The highly ranked countries were more or less expected with a few anomalies whereas the low ranked countries were expected to be the ones which are conflict-ridden, economically poor or have unstable governance. Afghanistan, Democratic Republic of Congo, and Cambodia do fit into those categories.

The ranking process is a helpful addition to the regression model by providing a more intuitive perspective on country performance, however, the low explanatory power of the model implies that some factors are excluded from the analysis.

Additionally, other important drivers that can influence the outcome of the pandemic are excluded from the scope of this research. Variations in the timing and nature of policies implemented by each country, differences in people's behavior and culture, and their level of interconnectedness can greatly affect their ability to cope with the crisis. Overall, while the PRS presents a valuable and statistically sound way to assess pandemic preparedness, it needs to be considered within the wider context of other contributing factors.

In conclusion, the integration of regression findings with country rankings reinforces the central argument of the study: **pandemic resilience is a multidimensional and context-dependent phenomenon.**

7.2 Contributions of this Study

This study contributes to the existing literature on global pandemic preparedness by providing a detailed and multidimensional analysis of the factors influencing excess mortality across 136 countries during 2020. Unlike many studies that examine isolated determinants i.e. either healthcare infrastructure or institutional quality/ trust in governance, this research integrates health system capacity, institutional quality, demographic structure, and broader socio-economic characteristics within a single empirical framework. This creates emphasis on the distinction between perceived preparedness and actual pandemic performance.

Many studies had called for an emphasis on social determinants along with other obvious factors to evaluate pandemic outcomes. Thus, this study contributes to the same by including all possible aspects, keeping in mind the data constraints. It also diverts the discussion towards maximizing the effectiveness of strategies as a combination of improved quality of indicators rather than focusing on one or the other. Another research gap addressed by this study, is the use of a reliable pandemic outcome measure which is Excess Mortality Rate.

The study also contributes by offering a more nuanced interpretation of commonly used structural variables like internet penetration and population density which are often treated as

straightforward indicators in empirical models; however, the findings of this research suggest that they reflect broader underlying dynamics, including data transparency, urbanization, and infrastructure availability. This highlights the need for careful interpretation of such variables and contributes to a more refined understanding of cross-country differences in pandemic outcomes.

This study goes an extra mile by ranking countries based on the Pandemic Resilience Score which not only provides further insight into future performance but also guides us to form targeted policies for each based on their weaknesses and strengths.

At a broader level, the study contributes by bridging the gap between theoretical discussions of pandemic preparedness and empirical evidence. It provides policy-relevant insights by identifying key areas like healthcare investment, governance quality, and protection of vulnerable populations—that are associated with better outcomes. At the same time, the moderate explanatory power of the model highlights the inherent complexity of pandemic phenomena and points to the importance of factors like early detection of pathogens, rapid implementation of strategies and other important factors.

7.3 Limitations of this Study

The study faces several important limitations that should be considered when interpreting results. Firstly, we need to keep in mind that institutional changes, new variants, and policy learning after COVID-19 may alter how predictors affect outcomes in a future pandemic. The relationships derived in the first step of methodology are based on data from Covid-19 period and we assume stability in terms of how these variables affect the dependent variable. Governments might have changed their strategies and adopted new technologies which might not get captured completely by the variables considered in the study.

Additionally, the pandemic resilience score assumes that a future COVID-19–like pandemic will have similar transmission mechanisms, as SARS-CoV-2, which may not hold for a pathogen with very different characteristics. The next potential pathogen could have a different mode of transmission, higher fatality rates etc. Certain variables might not even be

relevant in that case. Thus, there is an implicit assumption in place of similar pathogen threat. This makes the score less generalizable and reduces its external validity.

Finally, the regression has only included a limited set of observable predictors(trust, internet, health spending, elderly population, population density, etc.) due to data availability constraints. Several relevant factors such as policy timing and stringency, vaccination coverage, healthcare workforce capacity, cultural norms, and behavioral responses are either difficult to quantify or lack consistent cross-country data. The less explanatory power of this model can be seen by the value of R-squared derived from analysis.

There are also, of course, varied data reporting standards in different countries thus making the available secondary data less reliable for analysis. This highlights the need for future research to work on incorporating dynamic data into the dataset.

8. Policy Recommendations

Policy formulation is an integral part of effective governance and it guides decision making and actions in any field. The process involves designing, developing and implementing policies to address respective issues. This study gives us insight into the gaps in global pandemic preparedness and now requires policies and strategies to be implemented that address these particular gaps. Some of the suggested policy recommendations are discussed as follows:

A critical priority is the need to **rebuild and sustain trust in government**, which emerged as a significant determinant of lower excess mortality. It is recommended for governments to fund effective, consistent, and participatory communication policies which will not merely be confined to information campaigns but extend to the development of a real-time dashboard which highlights the health data, policy measures, and results in a digestible form.

Involvement of local bodies, like NGOs and influential leaders in the area, could help organize the citizens at various phases of implementation especially the early critical window. Third party committees should be set up to assess the present pandemic preparedness plans and even learn lessons from how they panned out during COVID-19. Proper accountability should be shown by responsible bodies because a slight delay could result in worse outcomes. Post COVID numerous countries saw a decline in trust on governance in nations like Latin American and African countries. This can be solved only by withstanding transparency in all aspects, communicating effectively and pushing forward research and evidence based strategies so that the citizens form trust and show the same during an emergency. This would result in long-term resilience and effective implementation of strategies.

Another key area of focus is the **protection of vulnerable elderly populations**, given their strong association with higher mortality outcomes. It is critical for governments to go past the response level and incorporate age-appropriate health practices as part of their institutions. This involves making sure that priority is given to the elderly people in vaccination programs, availability of care kits to be used at home, and preference when accessing health care during emergencies. It is also necessary to improve the capacity of the primary health care

institutions through hiring experts on geriatrics as well as increasing the number of long-term care centers. In addition to that, the measures have to be incorporated into other social protection frameworks that offer continued assistance to elderly people. It is noteworthy that elderly people continue to suffer vulnerabilities as a result of the failure of policy prioritization following the pandemic period in some regions like Latin America and Eastern Europe.

The study also highlights the significance of **enhancing the digital health infrastructure** along with tackling any inequalities present in its framework. Quick, effective communication solves problems that delayed action could never, thus it is essential that every country have a developed digital connectivity system, internet access and mobile connectivity, especially in remote areas.

A developed digital system could then serve multiple purposes including emergency information dissemination, fight against misinformation and availability of reliable data. Experience from the pandemic shows that higher connectivity does not only lead to spreading more accurate information but also false news, thus affecting individuals' conduct. Perception of what is happening around them. To tackle the problem, governments should cooperate with telecom companies in providing free or discounted access to authentic information about healthcare, specifically in times of crisis.

An obvious but rather important suggestion is sustained investment in healthcare be it infrastructure or for research. Here the important word is sustained i.e. it should be a continuous process since adapting to newer situations requires constant innovation. There will always be requirements that would have to be met considering we have limited resources and that too is spread unevenly across the globe. Also, there should be special focus on primary healthcare systems since they are the first line of defense during an outbreak.

Another crucial policy recommendation is the establishment of a **global pandemic preparedness and response fund**, specifically designed to support low-income and conflict-affected countries. Most of those states do not have the fiscal capacity to mobilize resources for future outbreaks, especially when they are dealing with economic issues and other forms of internal instability. Consequently, they remain more vulnerable than ever to

health risks. An international fund, funded mainly by high-income nations and international organizations, can help them get access to resources to deal with the situation in case of another outbreak. The Pandemic Fund started in 2022 after the Covid-19 pandemic is a step forward however it lacks the characteristic of an immediate response fund. It is mostly for capacity building.

The fund needs to have proper structures that would ensure its management is effective, accountable, and transparent. It can help low-income countries allocate resources to enhance their public health system, build surveillance capacity, and prepare for emergencies. In light of the interdependence of the global community, it makes sense to work on improving the resilience of vulnerable regions against future pandemics.

Moreover, there is also a need for **improved global coordination and data-sharing mechanisms**. Indeed, the pandemic revealed several weaknesses related to global cooperation, especially when it comes to early warning about possible outbreaks and subsequent response. Improvements in the global surveillance systems, better information reporting standards, and enhanced capacity for information exchanges could allow for earlier warnings and improved cooperation on responding to potential threats.

Ultimately, there is a need for measures that would help reduce the impact of social disparities in terms of pandemics. In particular, investments in education, ensuring financial stability, and ensuring access to the basics can make people less susceptible and increase their resilience to such challenges. A more balanced approach to development would ensure that resilience would be equally distributed across all countries.

To sum up, the analysis presented above, allows suggesting a series of policy recommendations for improving global resilience to pandemics. They include the need for strengthened public trust in government institutions, measures aimed at protecting vulnerable populations, increased investment in IT and healthcare infrastructure, and improved global cooperation on the issue. Moreover, addressing the problem of financing and other limitations to building resilience in low-income and conflict-affected countries.

9. Further Scope of the Study

This study can be further expanded to incorporate more complex methodologies, dynamic and accurate data, and deep contextual analysis which might contribute even more comprehensively to global pandemic preparedness strategies and policy formulation.

Some of the variables could not be included in this research due to data availability constraints, despite the chance that they would be strong determinants of pandemic outcomes such as policy response variables (timing and stringency of lockdowns, testing strategies, vaccination coverage) and other cultural or behavioural factors. Including them could have considerably increased the explanatory power of this study and consequently provided us with a higher R-Squared value. If data collection for these variables or their proxies are done on a cross country level, this could enable a more deeper understanding of actual preparedness strategies. Thus, further research in this respect demands for better data availability mechanisms to capture social dimensions that are generally hard to quantify.

Another area to focus on would be the refinement of the Pandemic Resilience Score by incorporating more complex methodologies. This study used regression coefficients as weights, but there are also other approaches like the Principal Component Analysis method or other machine learning methods that could be used to validate or refine the index. These approaches would be most effective when more relevant variables are also part of the dataset. In the end, comparing different methodologies could strengthen the robustness of the results.

Analysing interaction effects between variables could also very easily be done as an extension to this particular study. This would provide a deeper understanding of how different factors reinforce or offset each other in determining resilience. For example, the interaction effect between health expenditure and trust in governance and its impact on pandemic outcome would be very interesting for the study.

Future research could also integrate scenario based forecasting and simulation models to get actionable insights into what strategies actually work well and the areas of improvement related to pandemic preparedness. Under different conditions and unprecedented characteristics of a pathogen, it would be important that there be emphasis on practically

relevant problems and combating them would require regular simulation models for evidence based decision making during the actual epidemic.

All of these extensions are capable of providing an enhanced understanding of existing literature as well as promises newer insights regarding global preparedness for a future health crisis.

10. Conclusion

This study set out to examine the determinants of pandemic outcomes and to develop a comprehensive measure of pandemic resilience across countries. In doing so, this paper combines elements of regression analysis and the development of Pandemic Resilience Score (PRS) which helps to identify key determinants of excess mortality and prepare for potential future health crises.

First, empirical results of this study have demonstrated that pandemic outcomes depend on healthcare capacity, quality of institutions, demographic composition of the country, and socio-economic characteristics. Despite the negative relationship between health spending per capita and excess mortality, the effect of this variable is limited. Financial resources are crucial, yet not sufficient enough to ensure pandemic resilience. Instead, efficiency of systems and institutions plays a more important role than financial resources. The results also show that among all independent variables, trust in governance represents one of the most significant predictors of excess mortality. Therefore, institutional performance and effectiveness are crucial for pandemic resilience and success in fighting health crises. Additionally, age distribution, especially prevalence of elderly people in the population, is another important predictor of resilience.

The construction of PRS for ranking countries in terms of their preparedness adds value by transforming statistical analysis into comparison. As the ranking shows, there is considerable variation between different countries, indicating that resilience is not uniformly distributed around the world and is definitely not something that should be taken lightly. The highest and

lowest ranking countries were more or less expected however the few anomalies like UK and Sweden give us something to discuss about.

Several variables determine a country's level of resilience. The multidimensional nature of the dataset has been very clearly stated in this paper however data availability constraints have limited the explanatory power of the model. By using excess mortality as the outcome variable, the study also addresses limitations associated with reported COVID-19 data and provides a more accurate representation of pandemic impact. This study also challenges simplistic assumptions about variables such as population density and digital connectivity, demonstrating that their effects are context-dependent and often mediated by broader institutional and infrastructural conditions.

The resilience level of a nation is a combination of how much resources are allocated towards it and how those resources are used for better pandemic outcomes i.e. emphasis should be on both capacity and collaboration. On the one hand, healthcare institutions ensure the availability of the required resources whereas the effective use of these resources depends on the system of governance and public trust. The other social and demographic factors aggravate or offset baseline risks. Consequently, understanding all these factors helps to comprehend the complexity of pandemics and design proper policies.

In terms of policy-making implications, there should be further development of healthcare institutions and governance systems. Moreover, the vulnerable population should be protected, especially elderly people. Furthermore, digital and informational systems should be enhanced since these measures help reduce baseline risks. As for global initiatives, it is necessary to create a financial mechanism to support countries with low-income levels and involved in conflicts. It is impossible to neglect this aspect because of the interconnection between different countries in the world in terms of health issues. In addition, such an approach can reduce further dissemination of pandemics in the world.

Although the study has made some valuable contributions, there have been some shortcomings noted as well. For instance, the cross-sectional data make it difficult to measure any changes dynamically, while the weak explanatory strength indicates that some factors such as policy timing, behavior, and culture are beyond the scope of the model. This calls for

further research in the future with respect to longitudinal data, policy, and interaction effects among other things.

In conclusion, this research shows that in order to build resilience, an integrative framework encompassing the dimensions of healthcare capability, institution effectiveness, demographic knowledge, and readiness is essential. With the Pandemic Resilience Score created for this research as well as the findings derived empirically, important implications can be drawn in order to guide future policy development. Given that the threat of future health emergencies still looms large, there remain many lessons that can be learned from this assessment.

11. References

- Maccaro, A., Audia, C., Stokes, K., Masud, H., Sekalala, S., Pecchia, L., & Piaggio, D. (2023). Pandemic Preparedness: A Scoping Review of Best and Worst Practices from COVID-19. *Healthcare (Basel, Switzerland)*, 11(18), 2572.
<https://doi.org/10.3390/healthcare11182572>
- Rieger, M. O., & Wang, M. (2022). Trust in Government Actions During the COVID-19 Crisis. *Social indicators research*, 159(3), 967–989.
<https://link.springer.com/article/10.1007/s11205-021-02772-x>
- University VI, Kampala International. (2026). Pandemic Preparedness Lessons from COVID-19. *International journal of business and management sciences*. 12. 52-58.
10.59298/IDOSR/JES/06/1215258
- Ferdouse, Rafsana, Health Policy Responses To Pandemic Preparedness And Resilience (January 15, 2026).
<http://dx.doi.org/10.2139/ssrn.6137070>
- M. Marani, G.G. Katul, W.K. Pan, & A.J. Parolari, Intensity and frequency of extreme novel epidemics, *Proc. Natl. Acad. Sci. U.S.A.* 118 (35) e2105482118,(2021)
<https://doi.org/10.1073/pnas.2105482118>

- Shokoohi, M. , Osooli, M. and Stranges, S. (2020). COVID-19 Pandemic: What Can the West Learn From the East?. *International Journal of Health Policy and Management*, 9(10), 436-438. doi: 10.34172/IJHPM.2020.85
- Ndumbe-Eyoh, S., Muzumdar, P., Betker, C., & Oickle, D. (2021). 'Back to better': amplifying health equity, and determinants of health perspectives during the COVID-19 pandemic. *Global health promotion*, 28(2), 7–16.
<https://doi.org/10.1177/17579759211000975>
- Bisht, R., Saharia, R. & Sarma, J. COVID-19 and the burden of ill-health: a double crisis of disruptions and inequalities. *J. Soc. Econ. Dev.* 23 (Suppl 2), 342–356 (2021).
<https://doi.org/10.1007/s40847-020-00117-x>
- Julian W. Tang, Miguela A. Caniza, Mike Dinn, Dominic E. Dwyer, Jean-Michel Heraud, Lance C. Jennings, Jen Kok, Kin On Kwok, Yuguo Li, Tze Ping Loh, Linsey C. Marr, Eva Megumi Nara, Nelun Perera, Reiko Saito, Carlos Santillan-Salas, Sheena Sullivan, Matt Warner, Aripuanã Watanabe, Sabeen Khurshid Zaidi; An exploration of the political, social, economic and cultural factors affecting how different global regions initially reacted to the COVID-19 pandemic. *Interface Focus* 6 April 2022; 12 (2): 20210079.
<https://doi.org/10.1098/rsfs.2021.0079>
- Chawla, S., & Chawla, S. (2021). Comparative Analysis of Susceptibility and Severity of COVID-19 in Countries from the Eastern and the Western World Till March '21. *Microbiology insights*, 14, 11786361211041367.
<https://doi.org/10.1177/11786361211041367>
- Wang Z, Li Y, Xu R and Yang H (2022) How culture orientation influences the COVID-19 pandemic: An empirical analysis. *Front. Psychol.* 13:899730.
<https://doi.org/10.3389/fpsyg.2022.899730>
- Mario Coccia, Preparedness of countries to face COVID-19 pandemic crisis: Strategic positioning and factors supporting effective strategies of prevention of pandemic threats, *Environmental Research*, Volume 203, 2022, 111678, ISSN 0013-9351
<https://doi.org/10.1016/j.envres.2021.111678>
- Williams, B.A., Jones, C.H., Welch, V. et al. Outlook of pandemic preparedness in a post-COVID-19 world. *npj Vaccines* 8, 178 (2023).
<https://doi.org/10.1038/s41541-023-00773-0>

- Kachali, H., Haavisto, I., Leskelä, R. L., Väljä, A., & Nuutinen, M. (2022). Are preparedness indices reflective of pandemic preparedness? A COVID-19 reality check. *International journal of disaster risk reduction : IJDRR*, 77, 103074. <https://doi.org/10.1016/j.ijdr.2022.103074>
- Jennings, W., Valgarðsson, V., McKay, L., Stoker, G., Mello, E., & Baniamin, H. M. (2023). Trust and vaccine hesitancy during the COVID-19 pandemic: A cross-national analysis. *Vaccine: X*, 14, 100299. <https://doi.org/10.1016/j.jvacx.2023.100299>
- Suhaib JS. Ahmad, Konstantinos Degiannis, Joseph Borucki, Sjaak Pouwels, David Laith Rawaf, Marion Head, Chun Hei Li, Rami Archid, Ahmed R. Ahmed, Anil Lala, Wasif Raza, Katie Mellor, Doerte Wichmann, Aristomenis Exadaktylos, The most influential COVID-19 articles: A systematic review, *New Microbes and New Infections*, Volume 52, 2023, 101094, ISSN 2052-2975, <https://doi.org/10.1016/j.nmni.2023.101094>.
- Oppenheim B, Gallivan M, Madhav NK, Brown N, Serhiyenko V, Wolfe ND, et al. Assessing global preparedness for the next pandemic: development and application of an Epidemic Preparedness Index. *BMJ Global Health*. 2019;4:e001157. <https://doi.org/10.1136/bmjgh-2018-001157>
- Kelly-Cirino CD, Nkengasong J, Kettler H, Tongio I, Gay-Andrieu F, Escadafal C, et al. Importance of diagnostics in epidemic and pandemic preparedness. *BMJ Global Health*. 2019;4:e001179. <https://doi.org/10.1136/bmjgh-2018-001179>
- Aavitsland, P., Aguilera, X., Al-Abri, S. S., Amani, V., Aramburu, C. C., Attia, T. A., Blumberg, L. H., Chittaganpitch, M., Le Duc, J. W., Li, D., Mokhtariazad, T., Moussif, M., Ojo, O. E., Okwo-Bele, J. M., Saito, T., Sall, A. A., Salter, M. W. A. P., Sohn, M., & Wieler, L. H. (2021). Functioning of the International Health Regulations during the COVID-19 pandemic. *Lancet (London, England)*, 398(10308), 1283–1287. [https://doi.org/10.1016/S0140-6736\(21\)01911-5](https://doi.org/10.1016/S0140-6736(21)01911-5)
- Gostin L. (2004). The International Health Regulations and beyond. *The Lancet. Infectious diseases*, 4(10), 606–607. [https://doi.org/10.1016/S1473-3099\(04\)01142-9](https://doi.org/10.1016/S1473-3099(04)01142-9)
- Headley TY, Kim S, Tozan Y (2025) Countries' progress towards Global Health Security (GHS) increased health systems resilience during the Coronavirus Disease-19 (COVID-19) pandemic: A difference-in-difference study of 191 countries. *PLOS Glob Public Health* 5(1): e0004051. <https://doi.org/10.1371/journal.pgph.0004051>

- Duong, D. B., King, A. J., Grépin, K. A., Hsu, L. Y., Lim, J. F., Phillips, C., Thai, T. T., Venkatachalam, I., Vogt, F., Yam, E. L. Y., Bazley, S., Chang, L. D., Flaugh, R., Nagle, B., Ponniah, J. D., Sun, P., Trad, N. K., & Berwick, D. M. (2022). Strengthening national capacities for pandemic preparedness: a cross-country analysis of COVID-19 cases and deaths. *Health policy and planning*, 37(1), 55–64. <https://doi.org/10.1093/heapol/czab122>
- Mohammadpour, M., Zarifinezhad, E., Ghanbarzadegan, A., Naderimanesh, K., Shaarbafchizadeh, N., & Bastani, P. (2021). Main Factors Affecting the Readiness and Responsiveness of Healthcare Systems during Epidemic Crises: A Scoping Review on Cases of SARS, MERS, and COVID-19. *Iranian journal of medical sciences*, 46(2), 81–92. <https://doi.org/10.30476/ijms.2020.87608.1801>
- <https://data.worldbank.org/indicator/IT.NET.USER.ZS?end=2024&start=2020&view=char>
- <https://data.who.int/dashboards/covid19/cases?n=c>
- <https://reliefweb.int/report/world/world-risk-annual-report-global-preparedness-health-emergencies-global-preparedness>

12. Appendix

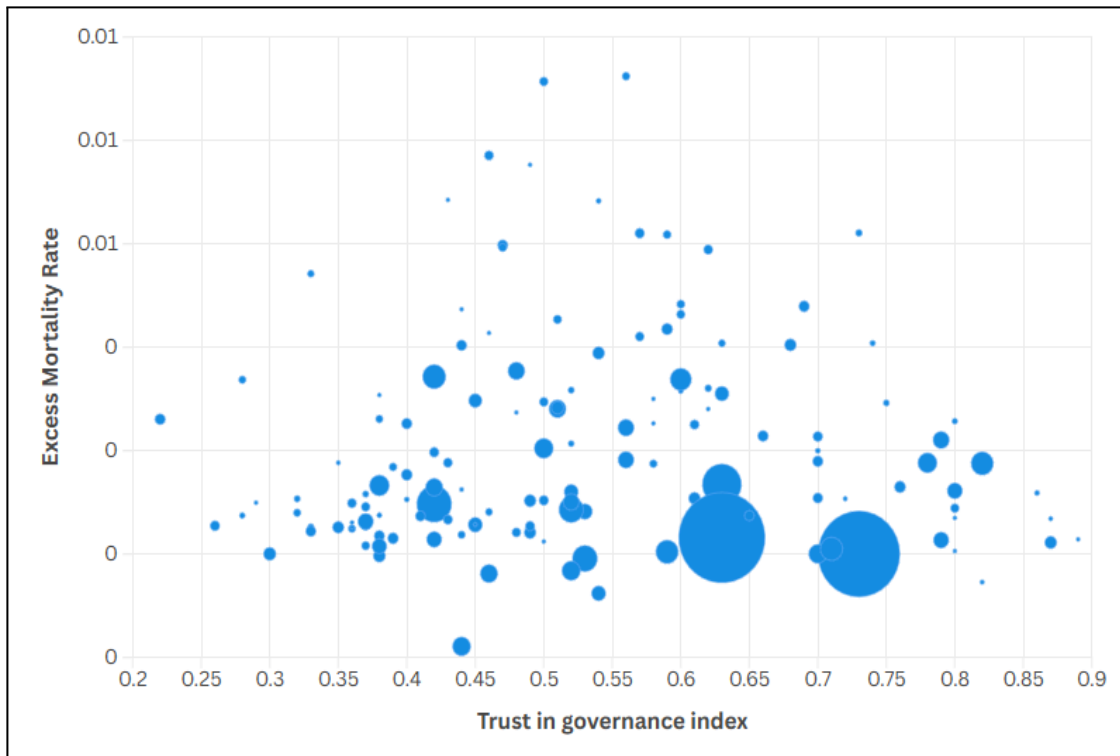


Figure-1

This bubble chart plots excess mortality rate (2020) on the vertical axis against trust in governance index on the horizontal axis, with each bubble representing a country and bubble size proportional to population. The clear negative relationship reveals that countries with higher trust in government consistently experienced substantially lower excess mortality during the first year of the COVID-19 pandemic, while low-trust nations cluster toward higher death rates. This visualization demonstrates trust in governance as a powerful social determinant of pandemic resilience, with large-population countries (larger bubbles) showing particularly pronounced effects.

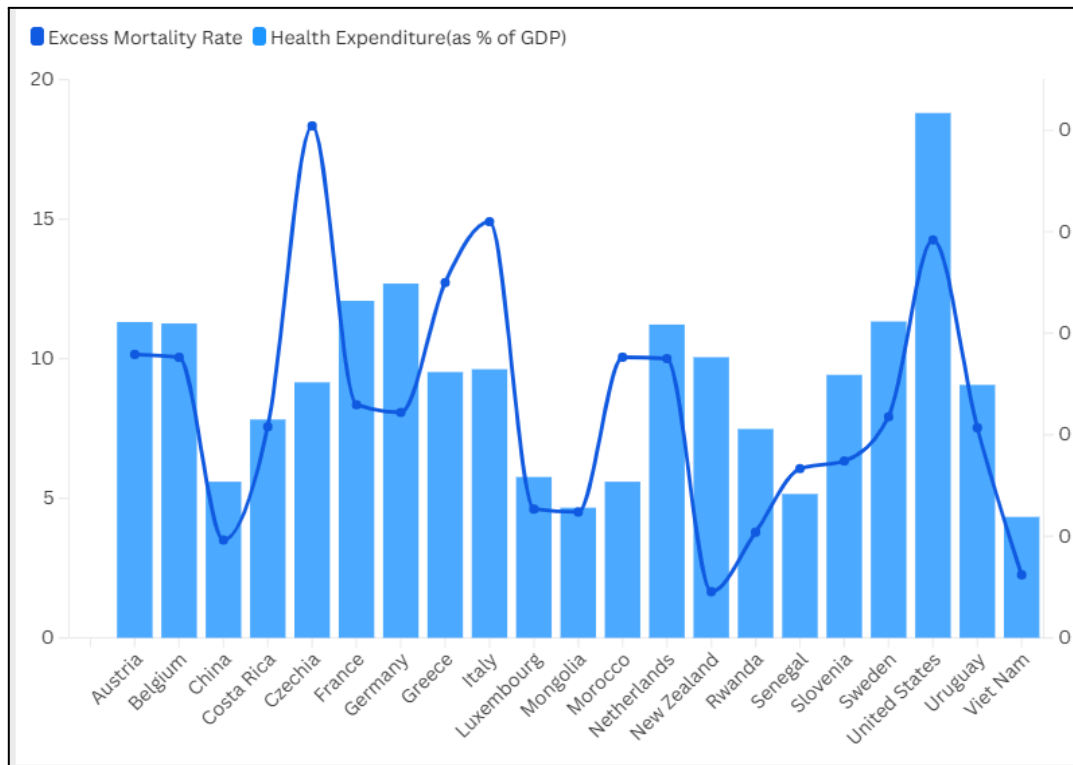


Figure-2

This dual-axis bar/line chart illustrates the non-linear relationship between health expenditure (% GDP) and excess mortality rate (2020) across 13 selected countries, demonstrating that higher spending did not systematically reduce mortality. The light blue bars show health expenditure levels (left axis), while the dark blue line plots corresponding excess mortality rates (right axis).

Notable examples include China and Costa Rica (low spending ~5%, low mortality), Mongolia and New Zealand (moderate spending, very low mortality), versus high spenders like Luxembourg (~11%, high mortality), Netherlands (~10%, moderate-high), Sweden (~11%, elevated), and United States (~17%, among highest mortality).

The irregular pattern confirms health expenditure alone was a poor predictor of pandemic outcomes.